

Newton-Cotes Cerradas

$$I \approx h \sum_{i=0}^n w_i f(a+ih)$$

$$h = \frac{b-a}{n}$$

Constantes para las fórmulas Cerradas de Newton – Cotes

n	α	i = 0	i = 1	i = 2	i = 3	i = 4	i = 5	i = 6	i = 7	i = 8	i = 9	i = 10
1	1/2	1	1									
2	1/3	1	4	1								
3	3/8	1	3	3	1							
4	2/45	7	32	12	32	7						
5	5/288	19	75	50	50	75	19					
6	1/140	41	216	27	272	27	216	41				
7	7/17280	751	3577	1323	2989	2989	1323	3577	751			
8	14/14175	989	5888	-928	10946	-4540	10946	-928	5888	989		
9	9/89600	2857	15741	1080	19344	5788	5788	19344	1080	15741	2857	
10	5/299376	16067	106300	-48525	272400	-260550	427368	-260550	272400	-48525	106300	16067

$$\int_0^1 (1-x^2) dx \quad n = 4$$